

Nova Financial Solutions

Predicting Price Moves with News Sentiment

10 Academy Week 1 Challenge - Final Report

May 2026

Publication-Style Analysis of Sentiment and Technical Indicators

Executive Summary

This report details the connection between financial news narratives and stock market performance. By analyzing over 1.4 million news articles and historical price data for major tech giants (AAPL, AMZN, GOOG, NVDA), we discovered a measurable positive correlation between news sentiment and daily stock returns.

Methodology

1. Exploratory Data Analysis (EDA): Understanding publication patterns.
2. Technical Analysis: Computing SMA, EMA, RSI, and MACD.
3. Correlation Analysis: Linking VADER sentiment scores to daily stock returns.

Correlation Analysis Results

Stock	Same-Day Corr	Lag Corr (Prev Day)
NVDA	0.2148	-0.0242
GOOG	0.1895	-0.0097
AMZN	0.1668	-0.2877
AAPL	0.1239	-0.2078

Key Insight:

The positive same-day correlation vs. negative/zero lag correlation suggests that markets are highly efficient. Sentiment is priced in almost immediately, leaving little predictive value for the next trading day.

Actionable Investment Strategies

- Sentiment-Filtered Breakouts: Use Sentiment > 0.2 to confirm SMA crossovers.
- The 'Hype' Warning: Be cautious if RSI > 70 and Sentiment is extremely high.
- Earnings Front-Running: Monitor sentiment 2-3 days before earnings calls.

Conclusion

Integrating qualitative sentiment with quantitative indicators provides a holistic view of market dynamics. At Nova Financial Solutions, we leverage this dual-approach to build robust predictive models.